

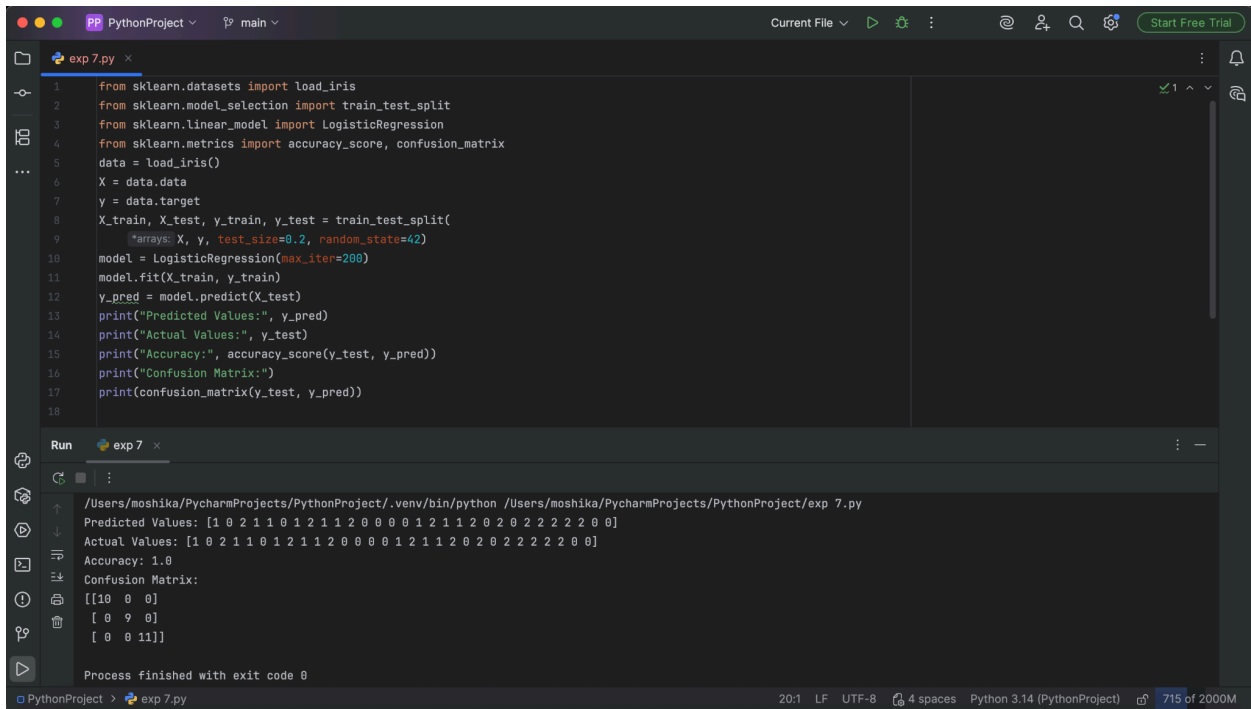
## EXPERIMENT - 7

**Aim:** To implement Logistic Regression in Python for classification of data into different classes.

### Procedure:

1. Import the required libraries.
2. Load the dataset.
3. Separate input features and target variables.
4. Split the data into training and testing sets.
5. Create a Logistic Regression model.
6. Train the model using training data.
7. Predict the class labels for test data.
8. Calculate the accuracy of the model.
9. Display the predicted results and accuracy.

### Output:



```
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.linear_model import LogisticRegression
4 from sklearn.metrics import accuracy_score, confusion_matrix
5 data = load_iris()
6 X = data.data
7 y = data.target
8 X_train, X_test, y_train, y_test = train_test_split(
9     *arrays: X, y, test_size=0.2, random_state=42)
10 model = LogisticRegression(max_iter=200)
11 model.fit(X_train, y_train)
12 y_pred = model.predict(X_test)
13 print("Predicted Values:", y_pred)
14 print("Actual Values:", y_test)
15 print("Accuracy:", accuracy_score(y_test, y_pred))
16 print("Confusion Matrix:")
17 print(confusion_matrix(y_test, y_pred))
```

Run exp 7 x

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 7.py
Predicted Values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0]
Actual Values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0]
Accuracy: 1.0
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Process finished with exit code 0

**Result:** The Logistic Regression model successfully classified the data and displayed the classification accuracy.